MILITARY SPECIFICATION

POTASSIUM CHLORIDE TABLETS

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Potassium Chloride Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Potassium Chloride Tablets and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

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S2. Classification. Shall be type XIII.

S5.3 The following is added to this paragraph:

Shall be potassium chloride in a 600 mg controlled - release tablet equivalent to 8 mEq of potassium.

Shall assay to contain not less than 540.0 and not more than 660.0 mg of potassium chloride when determined as specified in 4.2.1.

Cumulative release of potassium chloride shall comply with the following requirements when determined as specified in 4.2.2.

ELAPSED TIME	PERCENT KCl RELEASED
1 hr	35 - 55 percent
2 hrs	50 - 75 percent
4 hrs	75 - 105 percent
6 hrs	90 percent minimum

Tablets shall conform to identification tests for potassium and chloride when determined as specified in 4.2.4.

S6.4.1 tablets may be mottled.

3.2 Ingredients.

3.2.1 Potassium chloride. The potassium chloride used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Tests.

4.2.1 Assay of potassium chloride in the finished tablet.

Reagents and Solutions.

Nitric Acid, Concentrated.

Silver Nitrate, 0.1N Volumetric Solution, USP

Gelatin Solution, 1 percent
Dissolve 1 gram of Gelatin, USP, in 75 ml of boiling purified water. Cool and dilute to 100 ml with purified water.

Procedure:
Transfer 5 whole tablets equivalent to 3 grams of potassium chloride into a 250 ml volumetric flask containing 200 ml of purified water. Heat the solution to boiling, cool and dilute to volume with purified water. Mix well only after the solution has been diluted to volume. Filter a portion of the sample through suitable paper.

Pipet 20 ml of the filtrate into a 100 ml volumetric flask, and dilute to volume with purified water. Pipet 50 ml of this solution into a 250 ml beaker, add 50 ml of purified water, 5 ml of nitric acid, and 5 ml of 1 percent gelatin. Prepare a blank in a similar manner as the sample. Titrate the sample and blank solution potentiometrically with 0.1N silver nitrate using a silver electrode and a mercury-mercurous sulfate electrode.

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Calculation:

(V-Vb)N x 74.56 x 5 = mg potassium chloride/tablet

Where:

V = ml of silver nitrate consumed by sample
Vb = ml of silver nitrate consumed by blank
N = normality of silver nitrate

4.2.2 Potassium chloride cumulative release.

Atomic Absorption.

Reagents and Solutions.

Hydrochloric Acid, 0.6 normal (v/v).

Dilute 12 ml of concentrated hydrochloric acid to 200 ml with purified water.

Simulated Intestinal Fluid.

Dissolve 30 grams of sodium bicarbonate and 1.86 grams Pancreatin, NF, in sufficient purified water to make a total volume of 2000 ml.

Blank solution for intestinal fluid samples and standards.

Pipet 4.0 ml of Simulated Intestinal Fluid into a 100 ml volumetric flask and dilute to volume with 0.6 normal hydrochloric acid. Pipet 10.0 ml of this solution into a 100 ml volumetric flask, dilute to volume with 0.6 normal hydrochloric acid.

Potassium chloride in 0.6 normal hydrochloric acid standard solution.

Accurately weigh 2.0 grams of potassium chloride reference standard into a 1000 ml volumetric flask, dissolve in and dilute to volume with purified water. This is solution A. Into separate 100 ml volumetric flasks, pipet 1.0, 2.0, 3.0 and 4.0 ml of solution A and dilute to volume with 0.6 normal hydrochloric acid. Dilute 10 ml of each of these solutions to 100 ml with 0.6 normal hydrochloric acid. These are the standard solutions with concentrations of 0.2, 0.4, 0.6 and 0.8 mg/100 ml respectively.

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Potassium chloride in simulated intestinal fluid standard solution.

Into separate 100 ml volumetric flasks pipet 1.0, 2.0, 3.0 and 4.0 ml of potassium chloride solution A. Into each flask, pipet 4.0 ml of Simulated Intestinal Fluid and dilute to volume with 0.6 normal hydrochloric acid. Dilute 10.0 ml of each of these solutions to 100 ml with 0.6 normal hydrochloric acid. These are the standard solutions with concentrations of 0.2, 0.4, 0.6 and 0.8 mg/100 ml respectively.

Equipment

Sample Tube

A glass or plastic tube 80-100 mm long with an internal diameter of about 2.8 cm (as described in the British Pharmacopoeia 1968 edition), fitted at the lower end with 10 mesh stainless steel wire gauze (US 10 mesh).

Spectrophotometer

A suitable Atomic Absorption Spectrophotometer. Set the instrument for optimal operating conditions at 766.5 nm (potassium determination).

Procedure.

Transfer 250 ml of 0.6 normal hydrochloric acid into a 400 ml beaker and place in a water bath at 37° + 2°C. Place 5 tablets in the sample tube, attach the tube to a USP tablet disintegration tester and insert the tablets and tube into the 0.6 normal hydrochloric acid solution. On the upward stroke the gauze should just break the surface; on the downward stroke the upper rim of the tube should not go below the surface.

At the end of 1 hour, remove the beaker and replace it with another beaker containing 250 ml of 0.6 normal hydrochloric acid. At the end of the second hour, replace the beaker with one containing 250 ml of Simulated Intestinal Fluid. Do the same at the end of the third, fourth and fifth hours. At the end of 6 hours, remove the beaker and permit to cool to room temperature. These 6 beakers contain the sample solutions.

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First & second hour samples.

Pipet 5.0 ml of the first hour sample solution into a 200 ml volumetric flask and dilute to volume with 0.6 normal hydrochloric acid. Pipet 5.0 ml of the second hour sample solution into a 100 ml volumetric flask, and dilute to volume with 0.6 normal hydrochloric acid. Further dilute 4.0 ml of these solutions to 100 ml with 0.6 normal hydrochloric acid. Transfer approximately 30 ml of each solution to be measured into separate 50 ml beakers. Adjust the meter to 0 percent absorption with the coarse and fine minimum controls while aspirating 0.6 normal hydrochloric acid. Adjust the meter to 80 percent absorption with the maximum control while aspirating the 0.8 mg/100 ml potassium chloride standard in 0.6 normal hydrochloric acid.

Determine the percent absorbance of each of the standard solutions in 0.6 normal hydrochloric acid and of the first and second hour sample solutions, aspirating hydrochloric acid between each determination and noting the blank absorption which should remain at 0 percent.

Third through sixth hour samples.

Pipet 10.0 ml of the sample solution into a 100 ml volumetric flask and dilute to volume with 0.6 normal hydrochloric acid. Further dilute 4.0 ml of this solution to 100 ml with 0.6 normal hydrochloric acid. Transfer approximately 30 ml of each solution to be measured into separate 50 ml beakers. Adjust the meter to 0 percent absorption with the coarse and fine minimum controls while aspirating the Simulated Intestinal Fluid blank. Adjust the meter to 80 percent absorption with the maximum control while aspirating the 0.8 mg/100 ml potassium chloride standard in Simulated Intestinal Fluid.

Determine the percent absorbance of each of the standard solutions in Simulated Intestinal Fluid and the 3, 4, 5 and 6 hour sample solutions, aspirating the Simulated Intestinal Fluid blank between each determination and noting the blank absorption which should remain at 0 percent.

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Prepare separate calibration curves on linear graph paper for 0.6 percent hydrochloric acid and Simulated Intestinal Fluid by plotting potassium chloride concentration (mg/100 ml) of the standard solutions versus the corresponding percent absorbance. From the graphs, determine the mg/100 ml of potassium chloride corresponding to the sample percent absorbance for each sample solution.

Calculation:

First Hour:

$$W \times \frac{50,000}{A} = \text{percent Potassium Chloride Released}$$

Second Hour:

$$W \times \frac{25,000}{A} = \text{percent Potassium Chloride Released}$$

Third through Sixth Hours:

$$W \times \frac{12,500}{A} = \text{percent Potassium Chloride Released}$$

Where:

W = mg potassium chloride/100 ml of the sample from the graph.

A = mg potassium chloride/tablet from the assay.

4.2.3 Potassium chloride cumulative release. As an alternate the following selective electrode method may be employed.

Reagents and Solutions.

Hydrochloric Acid, 0.6 normal (v/v)

Dilute 12 ml of concentrated hydrochloric acid to 2000 ml with purified water.

Simulated Intestinal Fluid

Dissolve 30 grams of sodium bicarbonate and 1.86 grams of Pancreatin, NF, in sufficient purified water to make a total volume of 2000 ml.

Potassium Chloride in 0.6 normal Hydrochloric Acid Standard Solution.

Accurately weigh approximately 1400, 600, 240 and 120 mg of potassium chloride reference standard into separate 200 ml volumetric flasks and dilute to volume with 0.6 normal hydrochloric acid.

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Potassium Chloride in Simulated Intestinal Fluid Standard Solution.

Accurately weigh approximately 60, 120, 240 and 400 mg of potassium chloride into separate 200 ml volumetric flasks and dilute to volume with Simulated Intestinal Fluid. Filter these solutions through Whatman 2V filter paper.

Equipment.

Sample Tube.

A glass or plastic tube 80 to 100 mm long with an internal diameter of about 2.8 cm (as described in the British Pharmacopoeia 1968 edition) fitted at the lower end with 10 mesh stainless steel wire gauze (US 10 mesh).

pH Meter.

A pH meter capable of scale expansion in the millivolt mode, or a pH meter with a digital readout to 0.1 millivolt.

Electrodes.

Standard saturated calomel electrode with a salt bridge containing suitable salt bridge filling solution and a potassium specific electrode. The electrode and potassium detector shall be stored dried when not in use.

Procedure.

Place 250 ml of 0.6 normal hydrochloric acid in a 400 ml beaker and place it in a water bath at 37°C ± 2°C. Place 5 tablets in the sample tube, attach the tube to a USP tablet disintegration tester, and insert the tablets and the tube into the 0.6 normal hydrochloric acid solution. On the upward stroke the gauze should just break the surface; on the downward stroke the upper rim of the tube should not go below the surface.

At the end of one hour, remove the beaker and replace it with another beaker containing 250 ml of 0.6 normal hydrochloric acid. At the end of the second hour, replace the beaker with one containing 250 ml of Simulated Intestinal Fluid. Do the same at the end of the third, fourth and fifth hours. At the end of 6 hours, remove the beaker and let it cool to room temperature. Filter these solutions through Whatman 2V filter paper. These are the sample solutions.

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Starting with the lowest concentration potassium chloride standard solution in 0.6 normal hydrochloric acid, determine the potential of each of the standard solutions in 0.6 normal hydrochloric acid to the nearest 0.1 mv. Between each measurement place the electrodes in 0.6 normal hydrochloric acid for a few minutes, and blot them dry before placing them in the next solution. Determine the potentials of the 1 and 2 hour samples in the same manner as the standards. Determine the potentials of the potassium chloride standard solutions in Simulated Intestinal Fluid and all the remaining samples in the same manner as the previous samples. Do not leave the electrodes in intestinal fluid for more than a few minutes, because prolonged contact with this solution may poison the potassium electrode. To check for poisoning, redetermine the potential of the lower concentration potassium chloride standard solutions in Simulated Intestinal Fluid. If the potential is lower than it was before, the electrode has been poisoned, and should not be used.

Prepare separate calibration curves for 0.6 normal hydrochloric acid and Simulated Intestinal fluid by plotting on semilog graph paper the potentials (in millivolts) against the corresponding potassium chloride standard concentrations (in mg/ml). Determine the potassium chloride concentration for each sample from the graphs.

Calculation:

$$W \times \frac{50}{A} \times 100 = \text{percent Potassium Chloride Released}$$

Where:

W = mg potassium chloride/ml of the sample from the graph
A = mg potassium chloride/tablet from the assay

4.2.4 Identification.

Reduce two tablets to a fine powder, dissolve in 25 ml of purified water and filter through paper. The clear filtrate shall be used for both identities.

a) Potassium. Flame Test

Introduce a portion of the clear filtrate prepared above onto a platinum wire and place in the non-luminous flame of a Bunsen burner. View through a blue cobalt glass. A violet color shall be produced.

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b) Chloride. Reaction with Silver Nitrate

To the remainder of the filtrate prepared above, add silver nitrate TS. A white curdy precipitate, insoluble in nitric acid and soluble in a slight excess of ammonia TS, is formed.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issues of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing one hundred tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product container therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The closure shall be closure P (child-proof).

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred twenty (120) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred twenty (120) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. The label of the immediate container shall be a tri-part label consisting of the following:

1) a tab portion which shall remain attached to the immediate container. This portion shall bear:

   a) the National Drug Code (NDC)
   b) the lot or control number
   c) the expiration date

   Other label information may be included.

2) a tear-off portion adjacent to the tab portion or at the extreme end of the label bearing the following:

   a) the National Drug Code (NDC)
   b) the lot or control number
   c) the expiration date

   No other labeling shall appear thereon.
   This portion shall be easily removed for attachment to a prescription.

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3) the principal portion connected to the tear-off and/or the tab portion shall bear the following:

a) the National Stock Number
b) all other information required by the Federal Food, Drug, and Cosmetic Act.

The National Stock Number need not be preceded by any prefix.

This portion shall be easily removed.

Label adhesion test shall not apply to any portion of the tri-part.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE BELOW 86°F."

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE BELOW 86°F."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-008-5994	POTASSIUM CHLORIDE TABLETS, 0.6 Gram, 100s

Preparing activity:
DLA-DM

Project No. 6505-1941

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-37796(DLA-DM) POTASSIUM CHLORIDE TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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